

The empirical recurrence for OEIS A235780: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

7 September 2026

Abstract

OEIS A235780 records “Empirical recurrence of order 59”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 80$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 59, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 59 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A235780 is “Number of $(n+1) \times (2+1)$ 0..2 arrays with the difference of the upper and lower median value of each 2×2 subblock in lexicographically nondecreasing order columnwise and nonincreasing rowwise.”. It has offset 1 and begins

537, 7331, 86369, 924323, 9324103, 89825051, 837434835, 7600810347, ...

The entry states:

Empirical recurrence of order 59 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:49:05, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 222$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 80$ of the 222, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 80$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 444$ exact terms of the model returns order 59 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{59} c_i a(n-i),$$

c_1	24	c_{16}	1952361737177	c_{31}	308585864515806656	c_{46}	215273289599860736
c_2	−74	c_{17}	−12577163426633	c_{32}	−1008284481570055400	c_{47}	−742055937277165568
c_3	−2449	c_{18}	−10136398080492	c_{33}	−358985402417203624	c_{48}	56274977232715776
c_4	18595	c_{19}	114234408453777	c_{34}	2085327976863152248	c_{49}	219702228754628608
c_5	94708	c_{20}	11050568840451	c_{35}	43007904668849776	c_{50}	−51094941889462272
c_6	−1261598	c_{21}	−803446091129516	c_{36}	−3351917542314603168	c_{51}	−39421374433853440
c_7	−1071443	c_{22}	349089920354553	c_{37}	826293517071619808	c_{52}	14926317576257536
c_8	47292037	c_{23}	4422015391959621	c_{38}	4137274195102356224	c_{53}	3501135098806272
c_9	−47826516	c_{24}	−3870652858751583	c_{39}	−1985525353408364544	c_{54}	−2124262836011008
c_{10}	−1151788516	c_{25}	−19089151803957975	c_{40}	−3830349218598080256	c_{55}	−39662848573440
c_{11}	2554854469	c_{26}	24550652087467822	c_{41}	2729973335277492224	c_{56}	136694179299328
c_{12}	19462651214	c_{27}	64221556484495313	c_{42}	2545116016394030080	c_{57}	−11574399991808
c_{13}	−63665607028	c_{28}	−110998154551942362	c_{43}	−2543650125477748736	c_{58}	−2902324150272
c_{14}	−233446911005	c_{29}	−165090293067829792	c_{44}	−1100702082883457024	c_{59}	386547056640
c_{15}	1049507870008	c_{30}	380112379428521254	c_{45}	1654861731821764608		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 59, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $59 + 4$ terms removed returns order 59 as well, so $\deg q_{\min} = 59 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $59 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 59 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A235780.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.